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Module 1

Introduction to mobile computing, Middleware and Gateways, Application and services, Internet-Ubiquitous networks, Architecture and three-tier architecture for Mobile Computing, Design consideration for Mobile Computing.

1.Mobile Computing

The rapidly expanding technology of cellular communication, wireless LANs, and satellite services will make information accessible anywhere and at any time. Regardless of size, most mobile computers will be equipped with a wireless connection to the fixed part of the network, and, perhaps, to other mobile computers. The resulting computing environment, which is often referred to as mobile or nomadic computing, no longer requires users to maintain a fixed and universally known position in the network and enables almost unrestricted mobility. Mobility and portability will create an entire new class of applications and, possibly, new massive markets combining personal computing and consumer electronics.

A technology that allows transmission of data, via a computer, without having to be connected to a fixed physical link.

- Mobile voice communication is widely established throughout the world and has had a very rapid increase in the number of subscribers to the various cellular networks over the last few years. An extension of this technology is the ability to send and receive data across these cellular networks. This is the principle of mobile computing.

- Mobile data communication has become a very important and rapidly evolving technology as it allows users to transmit data from remote locations to other remote or fixed locations.

Mobile Computing is an umbrella term used to describe technologies that enable people to access network services anyplace, anytime, and anywhere. There are **2 aspects of mobile computing**

- – **User mobility:** users communicate “anytime, anywhere, with anyone” (example: read/write email on web browser). User mobility refers to a user who has access to the same or similar telecommunication services at different places, i.e., the user can be mobile, and the services will follow him or her.

- – **Device portability:** devices can be connected anytime, anywhere to the network. With device portability, the communication device moves. Many mechanisms in the network and inside the device have to make sure that communication is still possible while the device is moving.

Different names of Mobile Computing:-

VHE – Virtual Home Environment- VHE is defined as an environment in a foreign network such that the mobile users can experience the same computing experience as they have in their home or corporate computing environment.

Anywhere, anytime information- This is the generic definition of ubiquity, where the information is available anywhere all the time.

Nomadic computing- The computing environment is nomadic and moves along with the mobile user.

Pervasive computing- It is pervasive in nature and can be made available in any environment.

Ubiquitous computing- Ubiquitous computing is a concept in software engineering and computer science where computing is made to appear anytime and everywhere. In contrast to desktop computing, ubiquitous computing can occur using any device, in any location, and in any format.

Global service portability- Making a service portable and available in every environment.

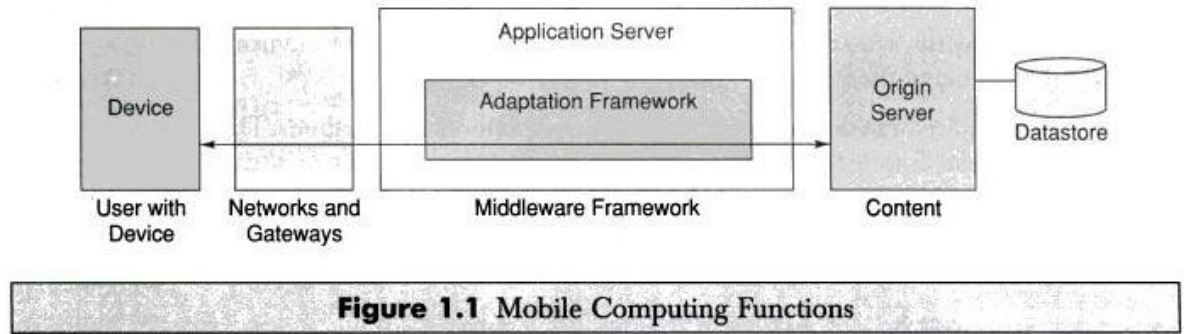
Wearable computers -Wearable computers, also known as wearables or body-borne computers, are small computing devices (nowadays usually electronic) that are worn under, with, or on top of clothing. The definition of 'wearable computer' may be narrow or broad, extending to smartphones or even ordinary wristwatches.

Mobile Computing Functions

We can define a computing environment as mobile if it supports one or more of the following characteristics:

- **User Mobility:** The user should be able to move from one physical location to another and use the same service. The service could be in a home or remote network. For example, a user moves from London to New York and uses Internet to access the corporate application the same way the user uses it in the home office.
- **Network Mobility:** Network mobility deals with two types of use-cases. In one use-case, the user is moving from one network to another and uses the same service seamlessly, An example could be a user moving from a WiFi network within the university campus and changing to 3G network outside while using the same online service.
- **Bearer Mobility:** The user should be able to move from one bearer to another and use the same service. An example could be a user using a service through WAP bearer in his home network in Bangalore. He moves to Coimbatore where WAP is not supported and switches over to the voice or SMS (short message service) bearer to access the same application.
- **Device Mobility:** The user should be able to move from one device to another and use the same service. An example could be sales representatives using their desktop computer in their home office. During the day while they are on the street they would like to use their Palmtop to access the application.
- **Session Mobility:** A user session should be able to move from one user-agent environment to another. An example could be a user using his service through a CDMA (Code Division Multiple Access) IX network. The user entered into the basement to park the car and got disconnected from his CDMA network. He goes to his home office and starts using the desktop. The unfinished session in the CDMA device moves from the mobile device to the desktop computer.
- **Agent Mobility:** The user-agent or the applications should be able to move from one node to another. Examples could be aglets, crawler software, or even a malicious worm or virus software that moves from one node to another. There is another use-case of mobile agent in the Cloud Computing paradigm, where applications will be moving from platform to platform and infrastructure to infrastructure depending on temporal and economic considerations. In Cloud Computing, there will not be any fixed association between the application and the host running it- software agents in the cloud will constantly be mobile.
- **Host Mobility:** The user device can be either a client or server. When it is a server or host, some of the complexities change. In case of host mobility, mobility of the IP needs to be taken care of.

The mobile computing functions can be logically divided into the following major segments (Fig. 1.1):



1. **User with device:** This means that this could be a fixed device like a desktop computer in an office or a portable device like mobile phone. Example: laptop computers, desktop computers, fixed telephone, mobile phones, digital TV with set-top box, palmtop computers, pocket PCs, two-way pagers, handheld terminals, etc.
2. **Network:** Whenever a user is mobile, he will use different networks at different locations at different times. Example: GSM, CDMA, iMode, Ethernet, Wireless LAN, Bluetooth, etc.
3. **Gateway:** This acts as an interface between different transport bearers. These gateways convert one specific transport bearer to another. Example: From a fixed phone [with voice interface] we access a service by pressing different keys on the telephone. These keys generate DTMF (Dual Tone Multi Frequency) signals. These analog signals are converted into digital data by the IVR (Interactive Voice Response) gateway to interface with a computer application. Other examples will be WAP gateway, SMS gateway, etc.
4. **Middleware:** This is more of a function rather than a separate visible node. In the present context, middleware handles the presentation and rendering of the content on a particular device. It may optionally also handle the security and personalization for different users.
5. **Content:** This is the domain where the origin server and content is. This could be an application, system, or even an aggregation of systems. The content can be mass market, personal or corporate content. The origin server will have some means of accessing the database and storage devices.

Mobile Computing Devices

The device for mobile computing can be either a computing or a communication device. In the computing device category it can be a desktop, laptop, or a palmtop computer. On the communication device side it can be a fixed Line telephone, a mobile telephone or a digital TV. Usage of these devices are becoming more and more integrated into a task flow where fixed and mobile, computing and communication functions are used together.

The device is a combination of hardware and software; the hardware is technically called the **User Equipment (UE)** with software inside, which functions as an agent to connect to the remote service- this software is called a **User Agent (UA)**. One of the most common UA today is a **Web browser**. When computing technology is embedded into equipment, Human-Computer Interaction (HCI) plays a critical role in effectiveness, efficiency, and user experience. This is particularly true as mobile information and communication devices are becoming smaller and more restricted with respect to

information presentation, data entry and dialogue control. The human computer interface challenges are:

- I. Interaction must be consistent from one device to another.
2. Interaction must be appropriate for the particular device and environment in which the system is being used.

2.MIDDLEWARE AND GATEWAYS

Any software layered between a user application and operating system is a middleware. Middleware examples are communication middleware, object-oriented middleware, message-oriented middleware, transaction processing middleware, database middleware, behavior management middleware, Remote Procedure Call (RPC) middleware, etc. There are some middleware components like behavior management middleware, which can be a layer between the client device and the application. In a mobile computing context we need different types of middleware components and gateways at different layers of the architecture (Fig. 1.2). These are:

1. Communication middleware.
2. Transaction processing middleware,
3. Behavior management middleware,
4. Communication gateways.

Communication Middleware

The application will communicate with different nodes and services through different communication middleware. Different connectors for different services will fall in this category. Examples could be TN3270 for IBM mainframe services, or Javamail connector for IMAP or POP3 services.

Communications Middleware is used in a variety of software systems, from mobile devices (PDAs, Android phones, iPads, etc.) to enterprise and database systems. The equipment in these systems varies in screen and visual display capabilities, bandwidth capacities, and processing power. **Communications Middleware can be cross-platform, that is, it facilitates communications between these differing devices. Communications Middleware can be cross-language, that is, it can understand and process multiple different operating languages (C, C++, Java, PHP, Ruby on Rails, etc.).** We can use a cell phone and a PC here as an example. They both function in vastly different capacities, but with Communications Middleware, are able to “talk” to and “work with” each other. This holds true for devices of similar capacities with different operating systems as well .

Transaction Processing Middleware

In many cases a service will offer **session-oriented dialogue (SoD)**. For a session we need to maintain a state over the stateless Internet. This is done through an application server. The user may be using a device, which demands a **sessionless dialogue (SID)** made of short sessionless transactions whereas the service at the backend offers a SoD. In such cases a separate middleware component will be required to convert a SoD to a SID. Management of the Web components will be handled by this middleware as well.

Behavior Management Middleware

Different devices deliver differently. We can have applications which are developed specially to deliver in a certain manner, For example, we can have one application for the Web, another for WAP, and a different one for SMS. On the contrary, we may choose to have a middleware, which will manage device- specific rendering at run-time. **This middleware will identify the device properly and handle all device-specific rendering independent of the application.** The system may be required to have some context awareness, which will be handled by the behavior management middleware.

Communication Gateways

Between the device and the middleware there will be a system of networks. Gateways are deployed when there are different transport bearers or networks with dissimilar protocols. For example, we need an IVR gateway to interface Voice with a computer, or a WAP gateway to access Internet over a mobile phone.

Figure 1.2 presents a schematic diagram of services in a mobile computing environment with different devices providing different services.

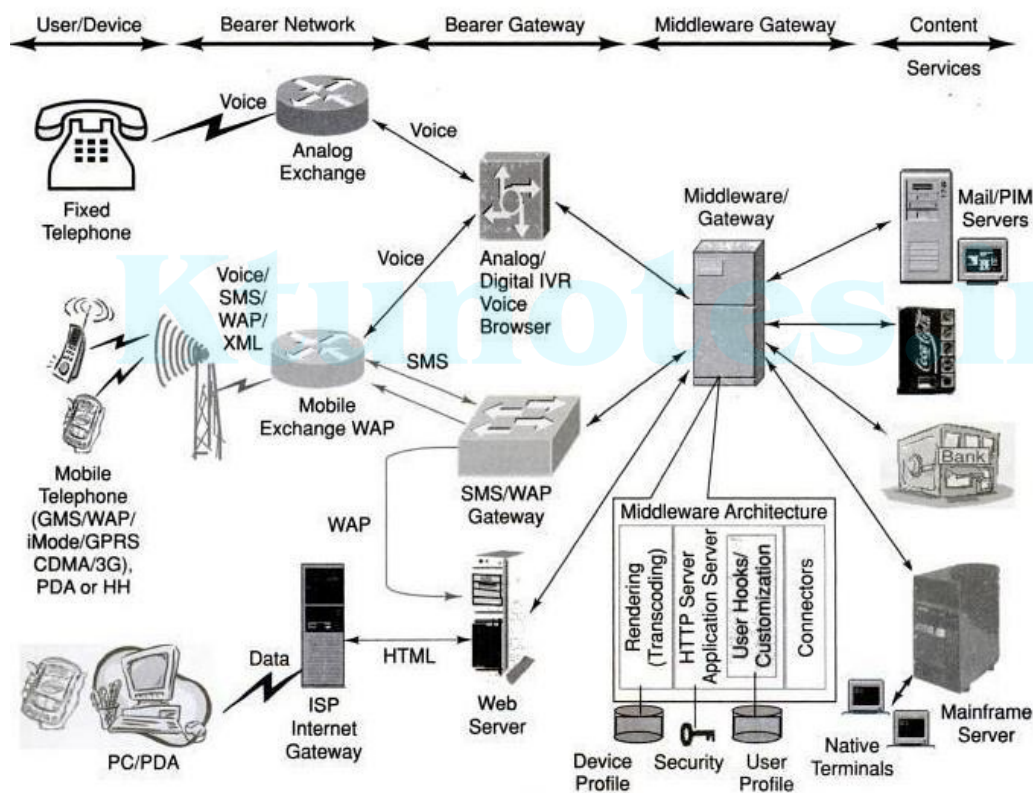


Figure 1.2 Schematic Representation of a Mobile Computing Environment

3.APPLICATION AND SERVICES

Data and information, through mobile computing services, are required by all people regardless of their mobility. Mobile users include people like mobile executives, sales people, service engineers, farmers in the field, milkmen, newspaper boys, courier or pizza delivery boy. Logically, everyone is a mobile user at some time or the other in life. For people who are stationary, mobile computing becomes necessary outside office hours. For example, we may need to do a bank transaction from home at night or respond to an urgent mail while at home.

There can be many applications and services for the mobile computing space. These applications or services run on the origin server. These are also known as content servers. Content will primarily be lifestyle-specific. An individual has different lifestyles in different social environments. Also, lifestyles change during the day. One individual can be an executive needing the corporate MIS (Management Information System) application during the day while at home the same individual can use applications related to lifestyle or entertainment. The list of possible mobile applications can never be complete. On the basis of life styles, they can be grouped into different categories, such as:

Personal: Belongs to the user (wallet, medical records, diary).

Perishable: Time-sensitive and of relevance and passes quickly (general news, breaking news, weather, sports, business news, stock quotes).

Transaction-oriented: Transactions need to be closed (bank transactions, utility bill payment, mobile shopping).

Location-specific: Information related to current geographical location (street direction map, restaurant guide).

Corporate: Corporate business information {mail, Enterprise Requirements Planning (ERP), inventory, directory, business alerts, reminders}.

Entertainment: Applications for fun, entertainment. Social networking sites like Facebook can be part of this category.

News: This is a very big basket of applications having different types of news. News could be political, current affairs, breaking news, business news, sports news, community news, etc. While people are on the move, they can always be connected to their culture and community through news, using mobile computing.

Youth: This is a very high growth market with different applications to suit the lifestyles of the youth. These are primarily message-based applications like person-to-person messaging, chat, forums, dating, etc.

Weather: There are different types of applications and services where mobile computing can make a difference. Notification services on weather is a very sought after application. If we have access to information related to the weather while on vacation or while driving from one location to another then the global positioning system (GPS) can help locate a person or sometimes save lives in case of a natural calamity.

Corporate application: Standard corporate information is an important piece of information for mobile workers and includes corporate mail, address book, appointments, MIS applications, corporate Intranet, corporate ERP, etc.

Sales force automation: This group will offer many applications. It will cater to the large population of sales personnel. Applications will include sales order bookings, inventory enquiry, shipment tracking, logistics related applications, etc. These applications will be very effective over wireless devices.

m-broker: Getting correct and timely information related to different stocks are very important. Also, online trading of stocks while on the move is quite critical for certain lifestyles. Stock tickers, stock alerts, stock quotes, and stock trading can be made ubiquitous so that users can check their portfolio and play an active role in the market.

Telebanking: We need to access our bank information for different transactions. Earlier, people used to go to the bank, but things are changing. Banks are coming to customers through telebanking. If telebanking can be made ubiquitous it helps the customer as well as the bank. Many banks in India are today offering banking over Internet (web), voice and mobile phones through SIPs.

m-shopping: This mobile application is used to shop with the help of mobile devices like Palm top, Pocket PC, mobile phones, etc. You can use this application to pay for a soft drink or soda from a vending machine in an airport or a movie theatre using a mobile phone, especially when you do not have sufficient cash.

Micropayment-based application: Micropayments involve transactions where the amount of money involved is not very high-it could be a maximum of Rs 1000 (\$ 25) or so. Micropayment through mobile phones can help rural people to do business effectively.

Interactive games: Many mobile network operators have started offering different types of contests and interactive games that can be played through mobile phones. The applications could be similar to any quiz, house, etc.

Interactive TV shows: Many TV companies around the world use email, SMS and Voice as a bearer for interactive TV or reality TV shows. In these shows viewers are encouraged to participate by asking questions, sharing opinions or even answering different quizzes. Nowadays viewers vote for their favorite TV stars using SMS.

Digital/Interactive TV: These are interactive TV programs through digital TV using set-top boxes and Internet. Video-on-demand, community programs, healthcare, and shopping applications are quite popular under this media category.

Experts on call: This is an application system for experts. Experts use these services to schedule their time and business with clients; clients use this to schedule business with the expert. A typical example could be to fix up an appointment with the tax consultant.

GPS-based systems: Applications related to location tracking come under this category. This could be a simple service like tracking a vehicle. Another example could be tracking an individual who got stuck due to bad weather while on a trekking trip. Fleet management companies and location-aware software systems need GPS-based applications.

Remote monitoring: This is important for children at home where parents monitor where their children are or what are they doing. Also, monitoring and controlling of home appliances will be part of this application.

Entertainment: This contains a very large basket of applications starting from horoscope to jokes. Many people in some part of Asia decide their day based on the planetary positions and horoscope information.

Directory services: This includes information related to movies, theatre, public telephones, restaurant guide, public information systems and Yellow pages.

Sports: This service offers online sports updates. In India live cricket score is the most popular mobile computing application. Getting scores of a live cricket match is the most popular mobile computer application. This service is available in India through Web, Voice, SMS, and WAP.

Maps/navigation guide: This is an application which has a lot of demand for traveling individuals. These services need to be location-aware to guide the user to use the most optimum path to reach a destination. The directions given by these applications also take traffic congestion, one way, etc., into consideration. GPS-based driving is becoming very popular in the US and advanced countries, where a user enters the postal address of the destination. The GPS-based system calculates the route, loads the right map and helps the driver navigate in real-time.

Virtual office: There are many people who are self-employed and do not have a physical office. Thus mobile and virtual office where they can check their mails, schedules, appointments, etc., while they are on the move are a must for them. Insurance agents and many other professions need these types of services.

m-exchange for industries: Manufacturing industry exchange from a mobile device can be a very cost effective solution for small/cottage industries. It may not be possible for a cottage industry to invest in a computer. However, accessing an exchange for a manufacturing company through a SMS may be affordable.

m-exchange for agricultural produce: Exchange for farmers on different type of agricultural products can be very useful for countries like India. If farmers can get information about where to get a good price for their product, it helps both farmers and consumers. There is a system unmo.edioupal.com to do exactly this. Think of this available over mobile phones.

Applications for speech/hearing challenged people: Telecommunication always meant communicating through Voice. There are people who cannot speak or hear. These include people with disabilities and senior citizens who lost their speech due to old age or after suffering a stroke. Text-based communication can help rehabilitate some of these disabled individuals.

Agricultural information: Think about a case where a farmer receives an alert in his local language through his mobile phone and immediately knows that the moisture content in air is 740/o. He can then decide how much to water his harvest. This can save his money, the harvest (excess water is sometimes harmful), and the scarce water resource. Portable devices with voice interface can change the economics of rural India with this kind of application.

Corporate knowledge-based applications: Many corporates today have a knowledge base. Making this ubiquitous can reduce cost and increase productivity.

Community knowledge-based applications: Knowledge is equally important for a community. Making knowledge ubiquitous always help society.

Distance learning: Applications related to distance learning are a must for countries with limited or no access to digital and information technology. For virtual schools in Asia or Africa, it is possible to have access to good faculty through the distance learning mode.

Digital library: These are libraries which can be accessed from anywhere anytime because of the Internet. Digital libraries can go a long way in shortening the digital divide as they also have support of local language and are easy and cheaper to commission.

Telemedicine and healthcare: Making telemedicine and healthcare easily available can save many lives. For example, a person complains of chest pain while traveling and requires immediate medical attention. He has to be taken to a doctor in a remote town. In this case, access to the patient's record can help expedite diagnosis. Reminder services for medicines or checkups can be very useful. In rural India, virtual clinics can help those who otherwise do not have access to medical care.

Micro-credit schemes: Micro-credit has a distinct role to play for a country's microeconomy. Grameen Bank with all its applications in Bangladesh is the best example of micro-credit.

Environmental protection and management: Ubiquity is a must for applications on environmental protection and management. Applications related to industrial hygiene will be part of this category. e-governance: These applications are very important to bridge the digital divide. The Bhoomi project of Karnataka government has computerized 20 million land records of 0.67 million farmers living in 30,000 villages in the state.

Virtual laboratories: There are many laboratories and knowledge repositories around the world which are made accessible to various cultures and countries through digital and information technology.

Community forums: There are different social and community meetings. In the case of India, panchayats can be made electronic. These may help increase the involvement of more people in community development work.

Law enforcements: Most of the time law enforcement staff are on the streets and need access to different types of services through wireless methods. These may be access to criminal records, information related to vehicles, or even a picture of the accident site taken through a MMS phone. This information can help insurance companies to resolve the claim faster.

Job facilitator: These could be either proactive alerts or information related to jobs and employment opportunities.

Telemetric applications: Almost every industry and sphere of life has the need for telemetry applications. Examples could be monitoring and control in manufacturing industry; vehicle tracking; meter reading; health care and emergency services; vending machine monitoring; research [telemetric orthodontic); control and service request for different emergency services for utilities like power plants, etc.

Downloads: Different types of downloads starting from ringtones to pictures are part of this category. In many countries this type of application is very popular. It is estimated that the market for ringtone downloads is more than 1 billion dollars.

Alerts and notifications: This can be either business or personal alerts. Simple examples could be breaking news alerts from a newspaper, Complex examples of alert could be for a doctor when the patient is in critical condition. In India many mobile operators are offering cricket alerts. In this service, subscribers receive score information every 15 minutes, about every wicket fall!

4. INTERNET-THE UBIQUITOUS NETWORK

For any content to be available anywhere, we need a ubiquitous network that will carry this content. As of today, there are two networks which are ubiquitous. One is the **telecommunication network** and the other is the **Internet network**. Both these networks are in real terms the network of networks. Different networks have been connected together using a common protocol (glue). In simple terms it can be stated that SS#7 is the glue for telecommunication network whereas TCP/IP is the glue for Internet. We need one of these networks to transport content from one place to another.

We have three types of basic content: audio, video and text. Some of these content can tolerate little delays in delivery whereas some cannot. **Packet switched networks** like Internet are better suited for content which can tolerate little delays. Telecommunication or **circuit switch networks** are better suited for real-time content that cannot tolerate delays. A ubiquitous application needs to use these networks to take the content from one place to another. network can be divided into three main segments, viz., **Core, Edge and Access**.

Core: As the name signifies, core is the **backbone of the network**, This is the innermost part of the network. The primary function of the core network is to deliver traffic efficiently at the least cost. Core looks at the traffic more from the bit stream point of view. Long-distance operators and backbone operators own core networks. This part of the network deals with transmission media and transfer points.

Edge/Distribution layer: As the name suggests, this is at the edge of the network. These are generally managed and owned by ISPs (Internet Service Providers) or local switches and exchanges. Edge looks at the traffic more from the service point of view. It is also responsible for the distribution of traffic.

Access: This part of the network services the end point or the device by which the service will be accessed. This deals with the last mile of transmission. This part is either through a wireline or the wireless. From the mobile computing point of view, this will be mostly through the wireless.

Core network and backbone network typically refer to the high capacity communication facilities that connect primary nodes. Core/backbone network provides path for the exchange of information between different sub-networks. Edge network provides information exchange between the access network and the core network. The devices and facilities in the edge networks are switches, routers, routing switches,

IADs and a variety of MAN/WAN devices, which are often called edge devices. Edge network provide entry points into carrier/service provider core/backbone networks. The Access layer is responsible for connecting devices to the network.

5.ARCHITECTURE FOR MOBILE COMPUTING

The network-centric mobile computing architecture uses three-tier architecture as shown in Figure 1.3. In the three-tier architecture, the first layer is the **User Interface or Presentation Tier**. This layer deals with user facing device handling and rendering. This tier includes a user system interface where user services (such as session, text input, dialog and display management} reside.

The second tier is the **Process Management or Application Tier**. This layer is for application programs or process management where business logic and rules are executed. This layer is capable of accommodating hundreds of users. In addition, the middle process management tier controls transactions and asynchronous queuing to ensure reliable completion of transactions.

The third and final tier is the **Database Management or Data Tier**. This layer is for database access and management. The three-tier architecture is better suited for an effective networked client/server design. It provides increased performance, flexibility, maintainability, reusability, and scalability, while hiding the complexity of distributed processing from the user.

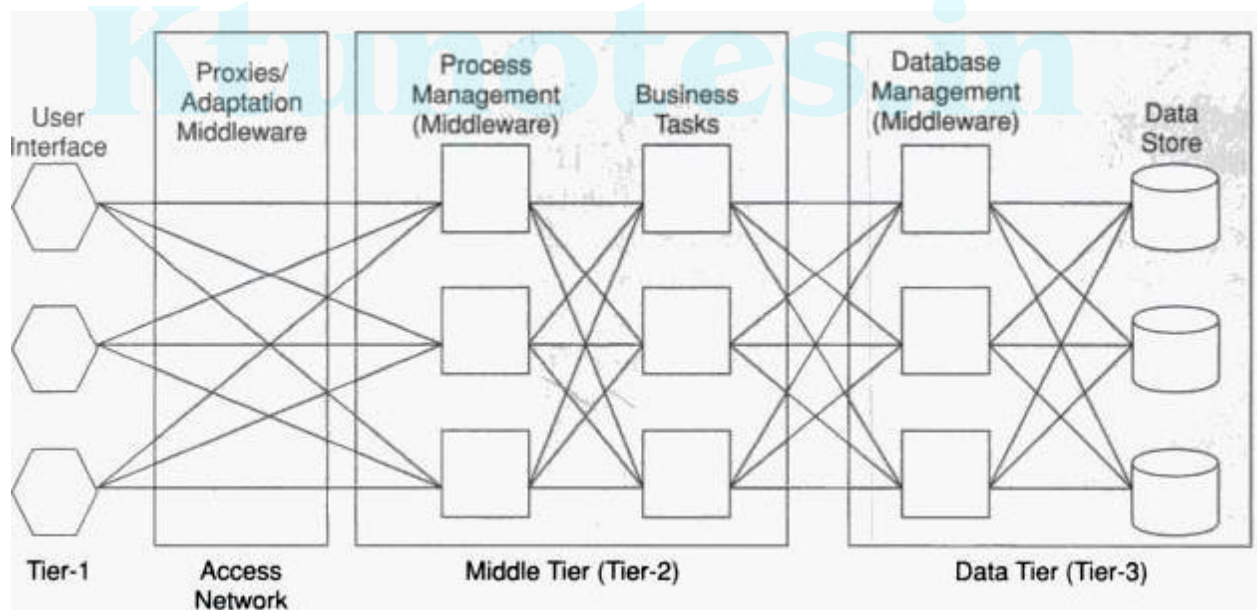


Fig:1.3 Three-tier Architecture for Mobile Computing

THREE-TIER ARCHITECTURE

To design a system for mobile computing, we need to keep in mind that the system will be used through any network, bearer, agent and device. To have universal access, it is desirable that the server is connected to a ubiquitous network like the Internet. To have access from any device, a web browser is desirable. The reason is simple; web browsers are ubiquitous, they are present in any computer. The browser agent can be Internet Explorer or Netscape Navigator or Mozilla or any other standard agent. Also, the system should preferably be context aware.

Figure 1.4 depicts a three-tier architecture for a mobile computing environment, These tiers are presentation tier, application tier and data tier. Depending upon the situation, these layers can be further sub layered.

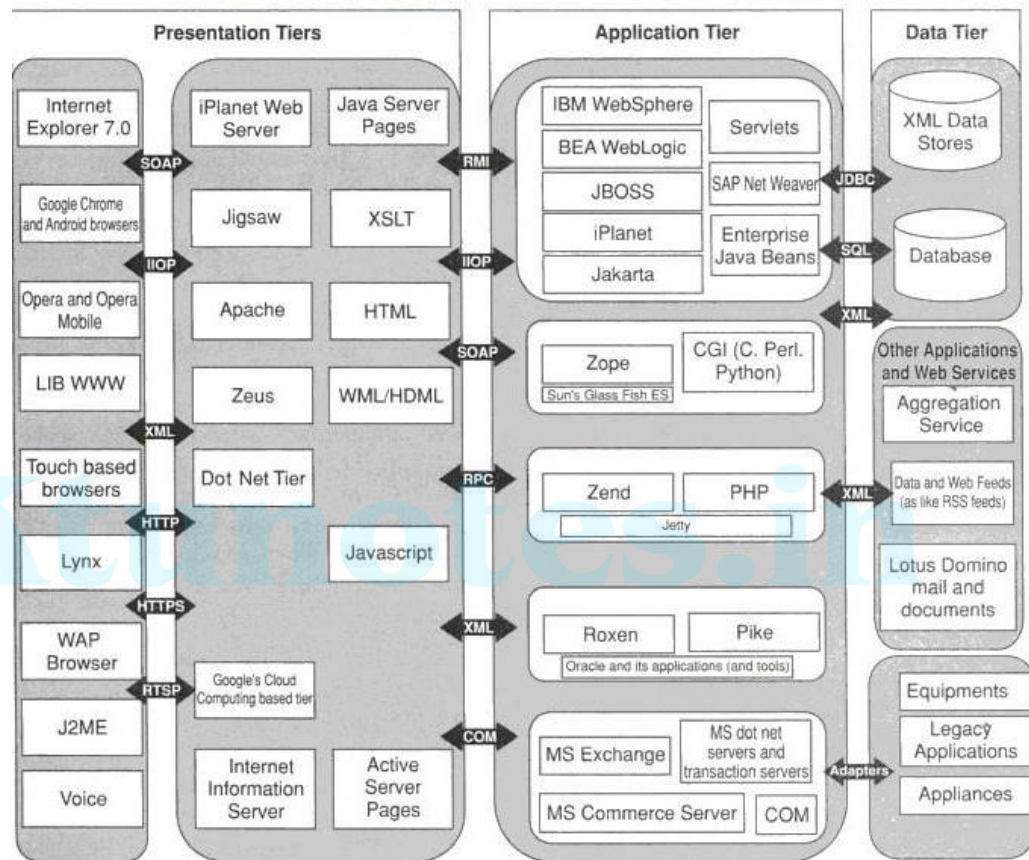


Fig:1.4 The Mobile Computing Architecture

Presentation(Tier-1)

This is the user facing system in the first tier. This is the layer of agent applications and systems. These applications run on the client device and offer all the user interfaces. This tier is responsible for presenting the information to the end user. Humans generally use visual and audio means to receive information from machines (with some exceptions like vibrator in mobile phones). Humans also use keyboard {laptop computers, cell phones), pen (tablet PC, palmtops), touch screen (kiosks), or Voice {telephone) to feed the data to the system. In the case of the visual, the presentation of information will be through a screen. Therefore, the visual presentation will relate to rendering on a screen.

'Presentation Tier' includes **web browsers** (like Mozilla, Lynx, Internet Explorer and Netscape Navigator), WAP browsers and customized client programs. **A mobile computing agent needs to be context-aware and device independent.**

In some cases, the agent software is an **applet** running on a browser or a virtual machine (Java Virtual Machine, for example]. The functions performed by these agent systems can range from relatively simple tasks like accessing some other application through HTTP API, to sophisticated applications like real time sales and inventory management across multiple vendors.

Some of these agents work as **web scrapers**. In a web scraper, the agent embeds functionality of the HTTP browser and functions like an automated web browser. The scraper picks up part of the data from the web page and filters off the remaining data according to some predefined template.

Application (Tier-2)

The application tier or middle tier is the "engine" of a ubiquitous application. It performs the **business logic** of processing user input, obtaining data, and making decisions. In certain cases, this layer will do the transcoding of data for appropriate rendering in the Presentation Tier. The Application Tier may include technology like **CGIs, Java, JSP, .NET services, PHP or ColdFusion**, deployed in products like **Apache, WebSphere, WebLogic, iPlanet, Pramati, JBOSS or ZENO**. The application tier is presentation and database-independent,

In a mobile computing environment, in addition to the business logic there are quite a few additional management functions that need to be performed. These functions relate to decisions on rendering, network management, security, datastore access, etc. Most of these functions are implemented using different middleware software. **A middleware framework is defined as a layer of software, which sits in the middle between the operating system and the user facing software.**

We can group middleware into the following major categories:

1. Message-oriented Middleware,
2. Transaction Processing Middleware.
3. Database Middleware,
4. Communication Middleware.
5. Distributed Object and Components.
6. Transcoding Middleware

Message-oriented Middleware (MOM)

Message-oriented Middleware is a middleware framework that loosely connects different applications through asynchronous exchange of messages. A MOM works over a networked environment without having to know what platform or processor the other application is resident on. The message can contain formatted data, requests for action, or unsolicited response. unsolicited response.

MOM is generally **asynchronous, peer-to-peer**, and works in **publish/ subscribe** fashion. In the publish/subscribe mode one or many objects subscribe to an event. As the event occurs, it will be published by the loosely coupled asynchronous object. The MOM will notify the subscribers about this event. However, most implementations of MOM support synchronous (request/response) message passing as well. MOM is most appropriate for event driven applications. When an event occurs, the publisher application hands on to the messaging middleware application the responsibility of notifying subscribers that the event has happened. In a net-centric environment, it can work as the integration platform for different applications. An example of MOM is Message Queue from IBM known as MQSeries. The equivalent from Java is JMS (Java Message Service).

Transaction Processing (TP) Middleware

Transaction Processing Middleware provides tools and an environment for developing transaction based distributed applications. An ideal TP system will be able to input data into the system at the point of information source and the output of the system is delivered at the point of information sink. In an ideal TP system, the device for input and output can potentially be different (Fig.1.5). Also, the output can be an unsolicited message for a device. **TP is used in data management, network access, security systems, delivery order processing, airline reservations, customer service, etc.,** to name a few. TP systems are generally capable of providing services to thousands of clients in a distributed client/server environment. CICS (Customer Information Control System) is one of the early TP application systems on IBM mainframe computers.

TP middleware maps numerous client requests through application-service routines to different application tasks. In addition to these processing tasks, **TP middleware includes numerous management features, such as restarting failed processes, dynamic load balancing and ensuring consistency of distributed data.** TP middleware is independent of the database architecture. TP middleware optimizes the use of resources by multiplexing many client functions on to a much smaller set of application-service routines. This also helps in reducing the response time. TP middleware provides a highly active system that includes services for delivery-order processing, terminal and forms management, data management, network access, authorization, and security. In the Java world and .NET systems, transaction processing is done through the J2EE application server with the help of net-centric and session beans. The help of entity beans is also used.

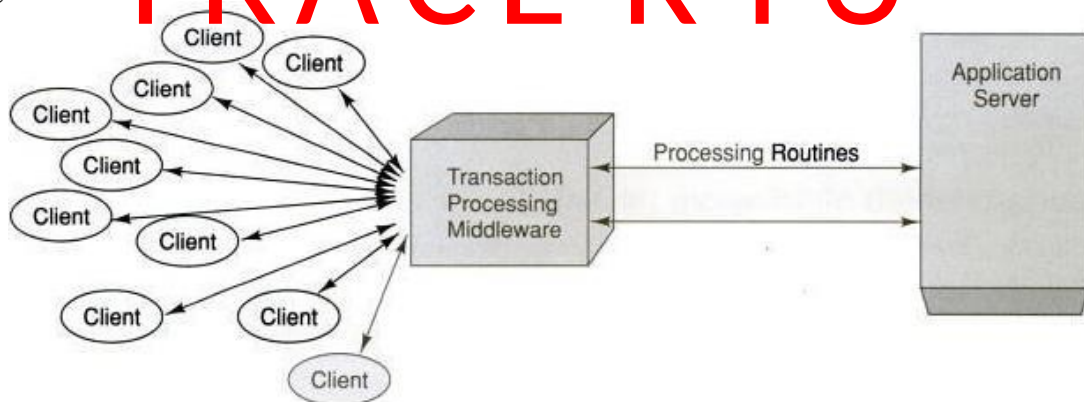


Fig:1.5 Transaction Processing Middleware

Communication Middleware

Communication Middleware is used to connect one application to another through some communication middleware, like connecting one application to another through telnet. These types of middleware are quite useful in the telecommunication world. There are many elements in the core telecommunication network where the user interface is through telnet. A mediation server automates the telnet protocol to communicate with these nodes in the network. Another example could be to integrate legacy applications through proprietary communication protocols like TN5250 or TN3270.

Distributed Object and Components

An example of distributed objects and components is CORBA (Common Object Request Broker Architecture). CORBA is an open distributed object computing infrastructure being standardized by the Object Management Group. CORBA simplifies many common network programming tasks used in a net-centric application environment. These are object registration, object location, and activation; request demultiplexing; framing and error-handling; parameter marshalling and unmarshalling; and operation dispatching. CORBA is vendor independent infrastructure. A CORBA-based program from any vendor on almost any computer, operating system, programming language and network, can interoperate with a CORBA-based program from the same or another vendor, on almost any other computer, operating system, programming language and network.

Transcoding Middleware

Transcoding Middleware is used to transcode one format of data to another to suit the need of the client. For example, if we want to access a web site through a mobile phone supporting WAP, we need to transcode the HTML page to WML page so that the mobile phone can access it. Another example could be accessing a map from a PDA. The same map, which can be shown in a computer, needs to be reduced in size to fit the PDA screen. Technically **transcoding is used for content adaptation to fit the need of the device. Content adaptation is also required to meet the network bandwidth needs.** For example, some frames in a video clip need to be dropped for a low bandwidth network. Content adaptation used to be done through proprietary protocols. To allow interoperability, IETF has accepted the Internet Content Adaptation Protocol (ICAP). ICAP is now standardized and described in RFC3507.

Internet Content Adaptation Protocol (ICAP)

ICAP, the Internet Content Adaptation Protocol, is a protocol aimed at providing simple object-based content vectoring for HTTP services. ICAP is a lightweight protocol to do transcoding on HTTP messages. This is similar to executing a "remote procedure call" on a HTTP request. The protocol allows ICAP clients to pass HTTP messages to ICAP servers for some sort of transformation. The server executes its transformation service on messages and sends back responses to the client, usually with modified messages. The adapted messages may be either HTTP requests or HTTP responses. ICAP works at the edge part of the network as depicted in figure 1.6.

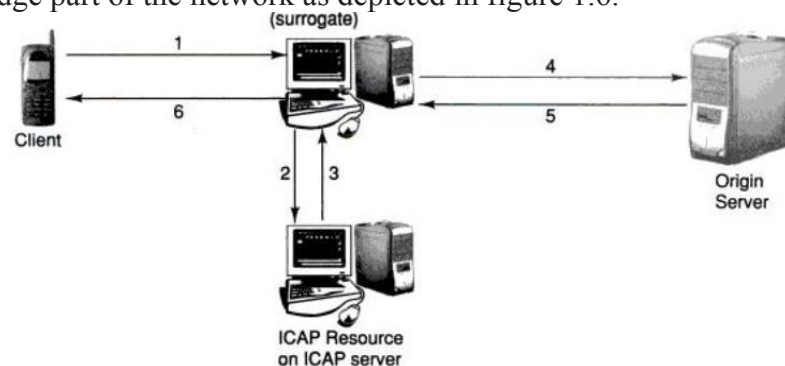


Fig:1.6 Typical data flow in an ICAP environment

Typical data flow in an ICAP environment is depicted in Figure 1.6 and described here.

1. A user agent makes a request to an ICAP-capable surrogate {ICAP client} for an object on an origin server.
2. The surrogate sends the request to the ICAP server.

3. The ICAP server executes the ICAP resource's service on the request and sends the possibly modified request, or a response to the request back to the ICAP client.
4. The surrogate sends the request, possibly different from the original client's request, to the origin server.
5. The origin server responds to the request.
6. The surrogate sends the reply (from either the ICAP or the origin server) to the client.

It is envisioned that in future, ICAP servers may be available to provide some of the following services:

- Suit content delivery based on network bandwidth.
- Suit content delivery based on device characteristics.
- Language translation based on the user's preference.
- Virus checking for the requested content.
- Content filtering based on sensor rating like PG (parental guidance), R (restricted).
- Local real-time advertisement insertion like television.
- Local real-time advertisement elimination for premium subscribers.
- Image magnification for the elderly.
- Image size reduction based on device display characteristics.

Web Services

Web services provide a standard means of communication and information exchange among different software applications, running on a variety of platforms or frameworks. Web service is a software system identified by a URI, whose public interfaces and bindings are defined using XML (eXtensible Markup Language). Its definition can be discovered by other software systems connected to the network, Using XML-based messages these systems may then interact with the Web service in a manner prescribed by its definition.

The basic architecture includes Web service technologies capable of:

- Exchanging messages.
- Describing Web services.
- Publishing and discovering Web service descriptions.

The Web services architecture defines the standards for exchange of messages between the service requester and service provider. Service providers are responsible for publishing a description of the services they provide. Requesters must be able to find and discover descriptions of the services.

Software agents in the basic architecture can take on one or all of the following roles:

- **Service requester**-requests the execution of a Web service.
- **Service provider**-processes a Web service request.
- **Discovery agency(Broker)**-agency through which a Web service description is published and made discoverable.

Data (Tier-3)

The Data Tier is used to store data needed by the application and acts as a repository for both temporary and permanent data. The data can be stored in any form of datastore or database. These can range from sophisticated relational database, legacy hierarchical database, to even simple text files.

Database Middleware

Database middleware allows the business logic to be independent and transparent of the database technology and the database vendor. Database middleware runs between the application program and the database. These are sometimes called database connectors as well. Examples of such middleware will be ODBC, JDBC, etc. Using these middleware, the application will be able to access data from any data source. Data sources can be text files, flat files, spreadsheets, or a network, relational, indexed, hierarchical, XML database, object database, etc., from vendors like Oracle, SQL, Sybase, etc.

6.DESIGN CONSIDERATIONS FOR MOBILE COMPUTING

The mobile computing environment needs to be context-independent as well as context-sensitive. **Context information is the information related to the surrounding environment of an actor in that environment. The term "context" means, all the information that helps determine the state of an object (or actor).** This object can be a person, a device, a place, a physical or computational object, the surrounding environment or any other entity being tracked by the system. In a mobile computing environment, context data is captured so that decisions can be made about how to adapt content or behavior to suit this context, Mobility implies that attributes associated with devices and users will change constantly. These changes mean that content and behavior of applications should be adapted to suit the current situation. There are many ways in which content and behavior can be adapted. Following are some examples:

1. Content with context awareness: Build each application with context awareness. There are different services for different devices. For example, a bank decides to offer mobile banking application through Internet, PDA and mobile phone using WAP. These services are different and are <http://www.mybank.com/inet.html>, <http://www.mybank.com/palm.html> and <http://www.mybank.com/wap.wml>, respectively. The service <http://www.mybank.com/inet.html> assumes that the user will use computers to access this service. Therefore it is safe to offer big pages with text boxes and drop down menus. Also, it is fine to add a few animated pictures for the new product the bank is launching. We know that <http://www.mybank.com/palm.html> is a service for a PalmOS PDA. As the display size is small, we design the screen to be compact for the PDA and do not offer the same product animation. For the WAP service at <http://www.mybank.com/wap.wml>, we do a completely different user interface; we make all drop down options available through the option button in the mobile phone and remove all the graphics and animations.

2. Content switch on context: Another way is to provide intelligence for the adaptation of content within the service. This adaptation happens transparent to the client. In this case the service is the same for Internet, PDA and WAP. All access the bank's service through <http://www.mybank.com/>. An intelligent piece of code identifies the agent to decide what type of device or context it is. This intelligent code does the adaptation at runtime based upon the agent in hand. The simplest way to do this is to look at the user-agent value at the HTTP header and decide whether to route the request to <http://mybank.com/inet.html> or <http://www.mybank.com/palm.html> or <http://www.mybank.com/wap.wml>

3. Content transcoding on context: Another way is to provide an underlying **middleware** platform that performs the adaptation of the content based on the context and behavior of the device. This adaptation happens transparent to the client and the application. The middleware platform is intelligent enough to identify the context either from the HTTP or additional customized parameters. In this case the service may be in html or XML, the middleware platform transcodes the code from XML to html, and wml on the fly.

Client Context Manager

A mobile computing application needs to operate in dynamic conditions. This is due to various device characteristics and network conditions. This demands a reactive platform that can make decisions about how to respond to changes to device capability, user preferences, enterprise policy, network policy and many other environmental factors. **Context can be used as the basis by which an adaptation manager or algorithm decides to modify content or application behavior.** We therefore need a **Client Context Manager** to gather and maintain information pertaining to the client device, user, network and the environment surrounding each mobile device.

All these information will be provided by a set of **Awareness Modules**. Awareness modules are sensors of various kinds. These **sensors** can be hardware sensors or software sensors or a combination of these. A hardware sensor can be used to identify the precise location of a user; whereas, a software sensor can be used to determine the type of the user agent. These awareness modules can be in the device, network, or even in the middleware. For example, an awareness module in the device will provide information about its capabilities. Another example could be a location manager that tracks the location and orientation of the mobile device.

Almost any information available at the time of an interaction can be seen as context information. Some examples are:

1. Identity: The device will be in a position to communicate its identity without any ambiguity.
2. Spatial information: Information related to the surrounding space. This relates to location, orientation, speed, elevation and acceleration.
3. Temporal information: Information related to time. This will be time of the day, date, time zone and season of the year.
4. Activity: This relates to the activity state of the user. This includes information like talking, reading, walking and running.
5. Availability of resources: This will relate to information about the device in use. This will include battery power, processing power etc.

The **mobile computing** environment is constrained in many ways. Mobile elements themselves are resource-poor and unreliable. Their network connectivity is often achieved through low-bandwidth wireless links.

These severe restrictions have a great impact on the design and structure of mobile computing applications and motivate the development of new computing models.

These **mobile computing models must provide** efficient access to both existing and new applications which is a key requirement for the wide acceptance of mobile computing.

An important design consideration is the type of functionality assigned to mobile hosts. Mobile units are still characterized as unreliable and prone to hard failures, i.e., theft, loss or accidental damage.

Thus following requirements must be considered for mobile computing environment:

1. Type of Application (Native or Mobile Web) :

- Before designing mobile application you need to determine type of application.
- If your application requires local processing. access to local resources than consider designing a native application.
- A native application is hard to maintain. requires separate distribution and upgrade infrastructure, are compatible only with target device/platform, requires more effort (sometimes huge) to port on different devices.
- A mobile web application is compatible with all devices with internet connection and a browser

2. Target device :

- Target device and platform (OS) plays a key role throughout design decisions making process.
- Design decisions are influenced by target device's screen size, resolution, orientations, memory, CPU performance characteristics. Operating systems capabilities, device hardware, user input mechanism (touch/non-touch), sensors (such as GPS or accelerometer etc.

3. User experience:

- User interface should be rich, intuitive and responsive. While using mobile application user is often distracted by external or internal (e.g. incoming call when user is in middle of a wizard) events. These events have to be factored in design.
- Screens should be identified with consideration that user is often focused on discrete individual task. Long data entry forms may irritate user and should ask for absolutely required data/information.

4. Resource Constraint:

- Every design decision should take into account the limited CPU, memory and battery life. Reading and writing to memory, wireless connections, specialized hardware, and processor speed all have an impact on the overall power usage.
- One cannot afford a long running process in application.
- In some cases, application processing should be offloaded from the device to avoid excessive resource consumption e.g. using notification or app directed SMS instead of polling to monitor a value/flag on server.

5. Multiple Platforms:

- There is high technology fragmentation in Mobile world. It is very unlikely that an application will target only one platform or only one device. In near future, requirement like same code base should support iPhone and iPad or Android Phone and Android tablet will arise. Architect should consider portability as important architectural goal.

- If you are developing for more than one device, design first for the subset of functionality that exists on all of the devices, and then customize for device-specific features when they are detected.

6. Security:

- Devices are more vulnerable than desktop, primarily due to lack of awareness. Moreover device can be lost easily. Ensure that the device – server communication is secured and server accepts request only from authentic source (device).
- If you are storing any confidential application or configuration data locally. ensure that the data is encrypted.

7. Network Communication:

- Network communication on device is slow and costly. Reduce network traffic by combining several commands in one request e.g. committing added, updated and deleted record in one request instead of firing separate request on each add/update/delete.
- Compress large text / XML data to lesser network traffic. Design for asynchronous communication.

Mobile elements are also resource poor relative to static hosts. For these reasons, there are approaches that treat the mobile unit as a dumb terminal running just a user-interface.

Besides the partition of data and computation of mobile applications between mobile and static elements, two important considerations in designing software system for mobile computing are: what is an appropriate model for mobile computing systems, and whether mobility should be made transparent to applications.